One Machine: the unified edition

OES end-to-end model, 28 October 2026. This edition puts the whole model in one text with one ledger. It brings together the verified build (layers 1–18), the integrated spine (layer 19), the canon synchronization (layer 20) and the sibling's 28 October ledger of joins (layer 21). It is a selected mathematical build replayed from matrices. It is not a source admission and not a change to the canon matrices.

AUTH = NOT_PASS · ScientificGold = FALSE · TIME_BIND = UNBOUND · DYN_BIND = NOT_INSTALLED · SCR6 = NOT_RUN_OPERAND_ABSENT · PredictionFreeze = NOT_TRIGGERED. No gate moves, and no term is added to the action.

Custody.

- Repository `sachinkaria24-byte/OES`, branch `claude/lagrangian-spectral-spacetime-model-t2253b`.
- `python verify.py` replays 21 layers in **412 checks**. Of these, 383 hold: THEOREM 268, OBSTRUCTION 40, ARITHMETIC 31, CONDITIONAL 17, UNFORMED 17, SELECTED_REFERENCE 8, REMOVED 2. The other 29 are REPORTED records.
- `python -m pytest` runs 31 tests.
- `ports/ts/` holds two drop-in audits for the sibling's app (35 rows, residual 0). `data/clocks.json` holds the explicit clock matrices, and `docs/LEDGER.md` holds the unified ledger.
- This text supersedes `one-machine-20261028-verified.pdf` (layers 1–18) and `one-machine-20261028-spine.pdf` (layers 19–20) as the current paper. Both stay in `docs/paper/` as history.

Abstract

The model has one datum, the signless incidence of K5. It has one action, fifteen oscillators with stiffness D^2, and one bath, the E8 root lattice. Everything else is a map between these three objects, a count, or a gate. This edition does three things.

It collapses the ledger. The sibling's 28 October ledger states about forty joins. The 129 sentences of the form "X contains every tick" or "X misses every nonzero tick" all carry one bit: the tick 864 lies in $\langle X \rangle$ exactly when 5 does not divide X. All 129 are checked against that one line, together with their step counts, their indices and 38 stated subgroup orders. The numbers that ride on the bit are shared integers, and no map joins them.

It puts every clock in one group. The normalizer of the Coxeter clock in W(E8) has order 240. In it:

- the tick is the commutator of the day clock and the Coxeter clock: $d\, c'\, d^{-1}\, c'^{-1} = c'^{6} = r\, r'$, two carrier rotations at once;
- the day–Coxeter group has order 120, centre $\langle c'^{5} \rangle$ and derived group the tick. It permutes its five day clocks as the affine group of the pentagon: the tick rotates them and the day doubles them, pentagon to pentagram;
- the golden clock has order 20, and its fourth power is the tick;
- the circle's mirror $x \mapsto -x$ is an involution that acts on the carrier as the pentagon reflection, fixing three atoms;
- the carrier and its twin can be exchanged only by elements of order 4, 8, 12 or 20, never by an involution, and every exchange doubles their index-5 glue.

The order-24 day element, which the sibling's build reports as not constructed, is given explicitly.

It states what is physical and what is not.

- The energy splits into four conserved pieces, and the vertex and edge sectors never exchange energy.
- The 240 roots are D8 plus one half-spin orbit.
- The source oscillator, of frequency 2, is not a carrier mode.
- Compton and Bohr lengths, the photoelectric effect, the double slit and entanglement stay unformed. No operand is stored for them, and the action has no dimensionful constant.

0. The model on one page

object	definition	what is proved	layer
datum	signless incidence M of K_n (n × C(n,2))	dim ker M = n holds only at n = 5	1
carrier	D = [[0, M], [M^T, 0]] on $\ell^2(V \sqcup E)$, dim 15	spec D^2 = {0^5, 3^8, 8^2}, Tr D^2 = 40, Tr D^4 = 200	1
action	$S = \int \tfrac{1}{2}\lvert\dot\psi\rvert^2 - \tfrac{1}{2}\langle\psi, D^2\psi\rangle$	15 oscillators: 5 free, 8 at √3, 2 at 2√2; four conserved energies	1, 21
spectral triple	($\mathbb{C}^V \oplus \mathbb{C}^E$, $\ell^2(V \sqcup E)$, D, γ)	heat trace $5 + 8e^{-3\beta} + 2e^{-8\beta}$; Witten index −5	2, 5
observer	Kalman readouts of (ψ, $\dot\psi$)	dark sector P5 ⊕ P5; five channels complete it	5, 20
bath	E8 root lattice, Δ = 240 − A	band [0, 256]; 11 critical families, 243,600 points, Morse sum 0; group speed √30	3, 16, 20
walks	W_n = closed walks of length n	one shift operator of order 16 and degree 58; its 11 in-band roots are the 11 critical values	14, 16
native action	source q on P5 ⊕ P5, quadratic port W_AS, six-site clique	$\ddot q + 4q = 0$ at the origin; sharp quartic 40/9	4, 10
glue	E8 ⊃ A4 ⊕ A4′	index 5, glue x ↦ ±2x; no involution exchanges the two	11, 21
waves	Hodge complex on the K5 edges	sound (4) and light (6) are isochronous at √5; Agni = Curl^2	17
circle	Z/4320 = $2^5 \cdot 3^3 \cdot 5$	one lemma; the tower 36 × 4 × 3 × 2 × 5; the tick bit	19, 21
clocks	the normalizer of ⟨c′⟩ in W(E8), order 240	every circle unit realized; the tick = [day, Coxeter]	19, 21
curvature, cosmology	Lin–Lu–Yau; spectral action on $M^4 \times F$	discretely flat bath; G and Λ_{cc} only with external inputs (CONDITIONAL)	6, 7
gates	AUTH, TIME_BIND, DYN_BIND, PredictionFreeze, W_L, T3081	none moves	13

1. The datum, the action and the energy

The datum. For the complete graph K_n, the signless incidence M (vertex × edge) has dim ker M = n exactly when n = 5, for every n. This is the only input. The fifteen atoms are the 5

vertices and the 10 edges of K5.

The carrier and its action.

- $D = [[0, M], [M^T, 0]]$ is odd, with grading $\gamma = +1$ on vertices and -1 on edges.
- $D^2 = \mathrm{diag}(MM^T, M^TM)$, with $MM^T = 3I + J$ on the vertices. The spectrum is $\{0^5, 3^8, 8^2\}$, so $\mathrm{Tr}\, D^2 = 40$ and $\mathrm{Tr}\, D^4 = 200$. The kernel $\ker D = P5 \subset \ell^2(E)$ is the irreducible 5 of S5.
- The action $S = \int \tfrac{1}{2}|\dot\psi|^2 - \tfrac{1}{2}\langle\psi, D^2\psi\rangle$ has no other term. A relative constant between its two terms is a rescaling of t.
- Its Euler–Lagrange equation $\ddot\psi + D^2\psi = 0$ is solved in closed form sector by sector. The frequencies are 0, $\sqrt{3}$ and $2\sqrt{2}$, and their squared ratio 8/3 is not a rational square.

Four conserved energies (THEOREM, layers 5 and 21). D^2 is block diagonal in $V \oplus E$, so the vertex and edge sectors never exchange energy. Each spectral sector $\lambda \in \{0, 3, 8\}$ is also invariant. Together these give four conserved pieces.

The sibling's checked datum takes:

- vertex value $(1, -1, 0, 0, 0) + (1, 1, 1, 1, 1)$;
- edge value M^T applied to the contrast;
- velocity the alternating cycle on the edges 01, 12, 23, 03.

Its quadratic form is $64 = 40 + 6 + 18$, and its energy is

> 34 = 20 (uniform vertex mode) + 3 (vertex contrast) + 9 (edge image of the contrast) + 2 (the cycle).

The vertex energy $E_V = 23$ and the edge energy $E_E = 11$ stay constant along the exact motion. The ledger's "34 = 12 + 20 + 2" is the spectral grouping of the same four pieces, with $12 = 3 + 9$.

Decoupling, quantized (CONDITIONAL). Under canonical quantization of S, the oscillating modes evolve by $U_V(t) \otimes U_E(t)$. The carrier therefore never entangles vertices with edges; entanglement can only be present in the initial data. The ledger's "entanglement is not stored" becomes a sharper statement: the carrier cannot create it. The quantization is a choice the canon does not store.

Symmetries of the action (THEOREM).

- The graph reflection of the pentagon, $i \mapsto -i$ on the vertices, keeps D^2 and fixes three atoms (one vertex and two edges).
- The fifteen-atom involution $i \mapsto 14 - i$ (the ledger's "above and below") fixes one atom and has eigenspaces of ranks 8 and 7. It sends a vertex to an edge, so it does not keep D^2.

The harmonic quartic (THEOREM). The S5-invariant quartics on the contrast plane are spanned by $|x|^4$ and Σx_i^4. For the plane's own Laplacian, the harmonic line is $2|x|^4 - 5\Sigma x_i^4$. For the ambient Laplacian of $\mathbb{R}^5$, it is $3|x|^4 - 7\Sigma x_i^4$. Neither enters the action: coupling a quartic would add a dimensionful constant.

The source is not a carrier mode (OBSTRUCTION). The stored source equation is $\ddot q + 4q = 0$, with frequency 2. Since $4 \notin \mathrm{spec}\, D^2 = \{0, 3, 8\}$, the source oscillator is not one of the fifteen carrier modes.

2. The spectral triple and the observer

Spectral content (THEOREM, layer 2).

- The heat trace is $Z(\beta) = 5 + 8e^{-3\beta} + 2e^{-8\beta}$.
- The moments are $\mathrm{Tr}\, D^{2k} = 15, 40, 200, 1240, 8840$.
- The spectral zeta function is $\zeta_D(s) = 8\cdot 3^{-s} + 2\cdot 8^{-s}$.
- The spectral dimension tends to 0 in both the UV and the IR. The carrier has no emergent continuum dimension.

Observation (THEOREM, layers 5 and 20).

- A scalar readout of $x = (\psi, \dot{\psi})$ sees at most 6 of the 30 dimensions, two for each spectral line.
- The Witten index is $\mathrm{Tr}\, \gamma\, e^{-\beta D^2} = -5$.
- The dark sector, reachable by edge forcing but invisible to $D\psi$, is the native source space $P5 \oplus P5$ (10 dimensions).

Five fixed channels are necessary and sufficient to complete the observer. With $H = [D; L]$, the reconstruction R satisfies $RH = I_{15}$, and $P_0 = I - 11K/24 + K^2/24$ with $K = D^2$. Of the 252 five-edge sensor designs, 162 are complete and 90 have rank 4. Every complete design has determinant 1/162. The 72 optimal designs have minimum eigenvalue $(3 - \sqrt{5})/6$.

3. The bath

Roots (THEOREM, layers 3 and 21).

- The 240 roots of E8 are 112 integer roots, which form D8, and 128 half-integer roots, which form one W(D8)-orbit (a half-spin weight set). So 248 = so(16) + 128.
- The root frame is $\Sigma\alpha\alpha^T = 60I_8$, and the root graph is 56-regular.
- [W(E8) : W(D8)] = 135. This is the D8 family at the top of the band, and the half-turn's 256 fixed momenta are 1 + 120 + 135.
- The ledger's "SU(2) is not derived" stands.

The band and its critical points (THEOREM, layers 3, 14, 16). The symbol is $\varepsilon(k) = 240 - \Sigma \cos(\alpha\cdot k)$, with range [0, 256]. The critical census lists $\lambda = 240 - \varepsilon$, the number of points and the Morse index:

family	λ	points	index		family	λ	points	index
E8	240	1	0		A6+A1 (edge)	−185/16	69,120	6
E7+A1	16	120	1		A7 (edge)	−320/27	17,280	7
E6+A2	−3	2,240	2		A7+A1	−12	8,640	7
D5+A3	−8	15,120	3		A8	−12	1,920	8
A4+A4	−10	48,384	4		D8	−16	135	8
A5+A2+A1	−11	80,640	5					

- There are 243,600 points in all, and the Morse sum is $0 = \chi(T^8)$.
- The eleven values of λ are exactly the eleven in-band roots of the walk operator below, so the band geometry and the walk data are one statement.

The walks (THEOREM, layer 14).

- $W_0 \ldots W_{90}$ are exact from two independent engines, and $W_0 \ldots W_{600}$ are known modulo primes.
- Every recurrence with $(r+1)(d+2) \le 455$ is excluded.
- The annihilator is one operator L_{16}, of order 16 and degree 58, the same modulo four primes. Its characteristic polynomial has the eleven in-band roots, $(z+12)^2$ among them,

and five roots outside the band.
- Asymptotically, $W_n = (16/\pi^4)\cdot 240^n \cdot n^{-4}\cdot(1 - 2/n + (23/15)/n^2 + O(n^{-3}))$.

Speed and density.

- The long-wave dispersion is $\varepsilon = 30|k|^2 - (3/2)|k|^4 + (1/24)|k|^6 + \ldots$, so the sound speed is $\sqrt{30}$ (layer 3). This is also the bath's maximal group speed, as a supremum that is never attained: $|\nabla\varepsilon|^2 \le 120\varepsilon$, by the frame and Cauchy–Schwarz (layer 20).
- The nine positive critical energies have distinct square-free parts, so their frequencies are independent over $\mathbb{Q}$ and their flow is dense on T^9 (layer 18).

Open. The convexity collar (223.58, 224) remains open.

4. The carrier inside the bath

Native action (layers 4 and 10).

- The source $q = (a, b)$ lives in $P5 \oplus P5$, with mass $M_s = I/4$.
- The port $g = W_{AS}(a \wedge b)$ is quadratic onto the six-dimensional cycle space L6, with $W_{AS}^T W_{AS} = (10/9)P$ and sharp quartic constant 40/9.
- The six-site clique has $A_{bb} = 241I - J$ and exterior coupling $BB^* = 183P_c + 495P_u$.
- At linear order the source obeys $\ddot{q} + 4q = 0$ and has no linear channel into the bath. Its frequency-4 harmonic lies inside the band through a thin channel.
- The printed chart T_6 preserves the contrast lattice, and $T_6|V_c$ has determinant 31, so $155 = 5 \times 31$.

Glue (THEOREM, layers 11 and 21). The carrier's own root system $A4 = \{e_i - e_j : i, j \le 5\}$ sits in E8, and its orthogonal complement is a twin A4′. The index is $[E8 : A4 \oplus A4'] = 5$, and the glue is $x \mapsto \pm 2x$. Layer 21 completes the picture with the stabilizer of the pair {A4, A4′} in W(E8), which has 57,600 elements:

- the 28,800 elements that keep each carrier act on the glue Z/5 by ±1;
- the 28,800 elements that exchange the carrier and its twin act on the glue by ±2, the doubling;
- every exchange has order 4, 8, 12 or 20. **No involution exchanges the carrier and its twin.**

The day's quarter d^6 (§6) is one such exchange, with $(d^6)^2 = -I$. The two carriers cannot be swapped as a mirror pair; on the bath, "above and below" is a quarter-turn on the glue.

Waves (THEOREM, layer 17).

- On the K5 edges, sound is the four longitudinal modes and light is the six transverse cycle modes.
- $d_0 d_0^T + \mathrm{Curl}^2 = 5I$, so the two are isochronous at $\sqrt{5}$. $\mathrm{Agni} = 5I - G = \mathrm{Curl}^2$ is the light operator.
- The two light helicities are the icosahedral 3 and 3′ of A5.
- No S5- or A5-symmetric medium converts sound into light.

5. The circle and the tick bit

One lemma (THEOREM, layer 19). $Z/4320$ has one subgroup $\langle d \rangle$ for each divisor d of 4320, of order 4320/d. Containment is divisibility of the gcds, and meets and joins are lcms and gcds. Two consequences:

- **Step and order are dual.** The atlas is the step 36, of order 120. The moon is the step 120, of order 36.
- **The spine is a chain** through the Coxeter subgroup:

  $0 \subset \langle 864 \rangle$ (tick, order 5) $\subset \langle 432 \rangle$ (yuga, 10) $\subset \langle 144 \rangle$ (Coxeter, 30) $\subset \langle 36 \rangle$ (atlas, 120) $\subset$ Z/4320.

  The tower is therefore $36 \times 4 \times 3 \times 2 \times 5$, and every element has a unique address.

By the Chinese remainder theorem, $Z/4320 = Z/32 \times Z/27 \times Z/5$, with orthogonal idempotents 3105, 2080 and 3456. The Coxeter step splits as $144 \equiv 2160 + 1440 + 864$: half-turn plus sun plus tick.

The tick bit (THEOREM, layer 21). For every x, $864 \in \langle x \rangle \iff 5 \nmid x \iff 3456 \cdot x \not\equiv 0 \pmod{4320}$. Consequences:

- exactly 24 of the 48 subgroups hold the tick (the $\langle g \rangle$ with $g \mid 864$);
- a subgroup holds all five tick levels {0, 864, 1728, 2592, 3456} or only 0;
- every width x prime to 5 that divides 4320 satisfies $x \times \text{index} = 864$;
- 16 of the 24 have order divisible by 20;
- the bit is not additive: 2 and 3 hold the tick, and $2 + 3 = 5$ does not.

The ledger is the bit (THEOREM). The sibling's ledger holds 129 tick sentences from 35 joins:

- energy, telescopes, golden, observer loop, mirrors, W0 reverse, phase, charge, spin;
- heat, monopole, field, Einstein line, two 36s, sign tensor, Om, karma, japa, mantra, bubbles, horizon;
- above and below, tick bounds, gravity dots, K5 chain, levels, pada, SM time, cakra, smear, readout, alpha, Hubble, height.

Every one of them is the residue of a number mod 5. Their step counts (for example $27 \times 192 \equiv 864$) and indices (for example $864 = 27 \times \gcd(224, 4320)$) agree, as do 38 stated subgroup orders. The readings riding on the bit are counts (ARITHMETIC), for example:

- $6 + 15 + 15 = 21 + 15 = 36$, with no $18 + 18$;
- the wings $21 + 7 + 77 = 105$;
- the phase weights $98 + 62 = 160$;
- $30/36 = 5/6$;
- the monopole chain $\langle 120 \rangle \subset \langle 60 \rangle \subset \langle 30 \rangle \subset \langle 10 \rangle$, none of which holds a tick.

No map joins these numbers to the objects they name.

6. One group for every clock

The Coxeter clock (SELECTED_REFERENCE; its properties are THEOREMs, layer 19). Let $c = s_1 \cdots s_8$ be the product of the simple reflections of E8. It has order 30, characteristic polynomial Φ_{30}, and exponents equal to the units mod 30. It permutes the 240 roots in eight free orbits of 30, so it is a symmetry of the bath already in the model. The selected conjugate $c' = w\, c\, w^{-1}$ satisfies

> $\mathbf{c'^6 = r\, r'}$,

the K5 vertex rotation r on the carrier A4 together with the twin rotation r′ on A4′: the tick turns two carriers. The other prime-order powers are:

- the sun c'^{10}, which turns four triangles ($A2^4$);
- the half-turn $c'^{15} = -I$, which flips eight edges ($A1^8$).

Their product is $c' = (-I)\cdot c'^{10}\cdot c'^{6}$, which matches $144 \equiv 2160 + 1440 + 864$.

Fixed momenta (THEOREM, layers 19 and 20). On the Brillouin torus, c^k fixes $|\det(c^k - I)| = \Phi_n(1)^{8/\varphi(n)}$ momenta, where $n = 30/\gcd(k, 30)$. That gives 25, 81 and 256 at orders 5, 3 and 2, and 1 otherwise. By the gradient lemma these are critical points: the origin and 24 points of A4+A4, the origin and 80 of E6+A2, and all 256 half-periods.

The normalizer (THEOREM, layer 19). The normalizer N of $\langle c' \rangle$ in W(E8) has order $240 = 30 \times 8$. Every unit m mod 30 is realized as a twist $w\, c'\, w^{-1} = c'^m$, and the centralizer is $\langle c' \rangle$ itself. The circle's automorphisms of its Coxeter subgroup are therefore all realized on the bath. The table lists the named elements; each is in `data/clocks.json`.

element	circle	twist m	order	on the carrier
c' (Coxeter clock)	step 144	1	30	mixes A4 and A4′
$c'^6 = r\, r'$ (tick)	step 864	1	5	rotates the pentagon on each carrier
d (day clock)	step 180	7	24	$d^4 = c'^5$; d^6 exchanges A4 and A4′
v (golden clock)	step 216	11	20	v^4 = the tick
m (mirror)	$x \mapsto -x$	$29 = -1$	2	the pentagon reflection, three atoms fixed

The tick is a commutator (THEOREM, layer 21). For the day clock d (order 24, characteristic polynomial Φ_{24}, $d^4 = c'^5$),

> $d\, c'\, d^{-1}\, c'^{-1} = c'^6 = r\, r'$.

The twist by 7 that the integrated build stops at and the two-carrier tick are therefore one fact. Write $G = \langle c', d \rangle$. Then:

- G has order 120, not the 720 of the abstract product;
- its centre is $\langle c'^5 \rangle$ of order 6, the circle's shared subgroup $\langle 720 \rangle$;
- its derived group is the tick, of order 5, and its abelianization is cyclic of order 24, generated by d;
- the day acts on the tick by doubling: $d\,(r\, r')\, d^{-1} = (r\, r')^2$.

Five day clocks (THEOREM, layer 21). G has exactly five conjugates of $\langle d \rangle$. On them:

- the tick acts as $x \mapsto x + 1$;
- the day acts as $x \mapsto 2x$;
- the centre acts trivially.

So G modulo its centre is the affine group AGL(1, 5), the twenty affine maps of the pentagon. In it the rotation is the tick, and the doubling, which sends the pentagon to the pentagram, is the day. The ledger's statements "doubling and rotation give 20 affine maps" and "the stored doubling does not commute with the order-5 rotation" are this group, realized on the bath.

The golden clock (THEOREM, layer 21). The 24 elements of order 20 in N all have twist 11. The one with $v^2 = c'^3$ satisfies $v^4 = c'^6 = r\, r'$, so every fourth golden step is the tick. Three cyclic groups of order 20 obey the same law:

- $\langle 216 \rangle$ on the circle, where $4 \times 216 = 864$;
- $\langle v \rangle$ on the bath;

- B = [[1,1],[1,0]] modulo 5, where B has order 20 (the Pisano period) and B^4 is unipotent of order 5.

The affine 20 of the five day clocks is not cyclic. The two kinds of 20 are different groups.

The mirror (THEOREM, layer 21). The circle's negation $x \mapsto -x$ fixes only 0 and 2160. It swaps $864 \leftrightarrow 3456$, $1728 \leftrightarrow 2592$, and the golden steps $1080 \leftrightarrow 3240$. It is realized by the 30 elements of N with twist -1, each of which inverts the clock, the tick and the sun. Ten of them are involutions that keep the carrier. On the carrier five act as a pentagon reflection, fixing one vertex and two edges, and five as its negative. The ledger's "graph reflection" and "negation on the circle" are one element of W(E8).

What can act on the bath (THEOREM, layer 19). A subgroup of the circle acts faithfully on the bath lattice exactly when its order is one of

> 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 18, 20, 24, 30.

This follows from explicit elements, the degree bound, a Sylow argument for 16, and the automorphism groups of A2 (for 36) and A4 (for 40 and 60). So:

- the maximal clocks are the Coxeter clock (30), the day (24), the golden clock (20) and $\langle 240 \rangle$ (18);
- the atlas, moon, nakshatra, pada and Human Design gate cannot act. On the circle the day and the Coxeter clock commute and generate the atlas, but on the bath no element of order 24 commutes with c'.

7. The canon and the sibling, reconciled

Position. The Quick Access workbook has a new top sheet, 00_FULL_RECONCILIATION_20261028, with review cutoff 17:19:41 UTC. It keeps OMSOON736 as the controlling numbered baseline. No new Matrix ID was issued, and every frozen gate stays. It still lists this repository's 14:37 build (18 layers, 320 checks) as the latest One Machine status. This edition arrived after that cutoff.

The Canon Check (layer 20).

- Its preparation identities are replayed: the Bernstein ellipse, $\|C\|^2 = 495$ and the optimal $\eta^* = 2\,\mathrm{arcosh}(d/8T)$. The horizon laws $\varepsilon^{-1/2}$ and $\varepsilon^{-1/3}$ stay CONDITIONAL on the inherited exterior contract.
- Its scalar bound $(r+1)(d+2) \geq 195$ follows from layer 14.

What the sibling adopted. At 17:06 the sibling's CANON_READ adopted the corrections of layer 20:

- "360000 is not a step" is removed;
- both harmonic quartics are named with their Laplacians;
- 5 and 25 are dual to the tick;
- the fixed determinants 25, 81 and 256 are stated;
- the day–Coxeter extension has order 120.

Its 18:17 full edition keeps these, and twice notes that "the order-24 element was not constructed". Layer 21 constructs it, and `data/clocks.json` and `ports/ts/ledgerJoins.ts` carry it into the sibling's app.

The separate model, replayed (layer 20). "Finite Spectral Dynamics with Complete Observation and Exact Cyclic Addresses" is rebuilt from its definitions:

- the observer completion and the sensor census (§2);
- the 25-coordinate flag geometry, with intersection determinant −48 and vertex distance exactly √(8/7);
- the collocated transfer functions $G_V = (z^2+7)/((z^2+3)(z^2+8))$ and $G_E = 1/(2z^2) + 2/(5(z^2+3)) + 1/(10(z^2+8))$.

8. The unified ledger

`docs/LEDGER.md` lists every join of both builds with its status and the layer that checks it. A test keeps it current. In short:

join	unified statement	status
circle lemma, tower	subgroups = divisors; 36 × 4 × 3 × 2 × 5	THEOREM
all tick joins	one bit: 5 ∤ X	THEOREM (the readings are ARITHMETIC)
rotor	$c'^6 = r\, r' = [d, c']$	THEOREM
golden subgroup	⟨216⟩, ⟨v⟩, B mod 5: the fourth power is the tick	THEOREM
oscillating tick	the day doubles its five conjugates; G/Z = AGL(1,5)	THEOREM
mirrors, W0 reverse	twist −1 involutions; the pentagon reflection on the carrier	THEOREM
above and below	no involutive exchange of carrier and twin; i ↦ 14 − i does not keep D^2	THEOREM
energy split	34 = 20 + (3 + 9) + 2; E_V and E_E conserved	THEOREM
observer	five channels; RH = I_{15}	THEOREM
spin counts	240 = D8 + one half-spin orbit; 135 = the band-top D8 family	THEOREM
field	$\ddot{q} + 4q = 0$ is not a carrier mode	OBSTRUCTION
sign tensor, entanglement	10 + 10 is a count; the carrier cannot entangle V with E	CONDITIONAL
charge, monopole, Einstein line, two 36s, Om, karma, bubbles, horizon	counts riding on the bit	ARITHMETIC
order-24 element	constructed, in `data/clocks.json`	THEOREM
source infinity, foam, double slit, photoelectric, electron, photon, proton, Compton, Bohr, Bell count, 19 SM parameters	no operand, no dimensionful constant	UNFORMED

9. What is not formed, and the gates

Not formed.

- The six SCR6 operands, the original seams, the 502- and 237-dimensional operators, the MPKG adapter, and the historical Agni, Soma, Suraj, Sawan and Om matrices.
- The Krishna lift's operands, the epoch and the ayanamsha.
- GKSL, SI units, BRST, BV and Einstein–Hilbert.

The carrier action is constant-free, so it can predict only ratios such as √(8/3). A length or a mass needs one authenticated dimensionful datum, which is TIME_BIND.

Gate	State	What closes it
AUTH / SCR6	NOT_PASS / operand absent	the owner's signature; the six SCR6 operands of a chosen physical child
TIME_BIND	UNBOUND	one authenticated clock datum (frequency ratios are calibration-invariant)
DYN_BIND	NOT_INSTALLED	the physical child, the generation multiplicity, the chirality index
PredictionFreeze	NOT_TRIGGERED	the owner freezes the register before new comparisons
W_L	NOT_PASS by evidence	the recorded PMNS candidate is excluded ($\chi^2 = 10.6$ on 2 dof)
T3081	UNOPENED	the official host must be allowed
ScientificGold	FALSE	a frozen prediction, independent data and a preregistered test

Open mathematics.

- The convexity collar (223.58, 224).
- The rational coefficients of L_{16}, and a proof that it annihilates W_n for every n.
- A preparation protocol matching the $\varepsilon^{-1/2}$ and $\varepsilon^{-1/3}$ bounds.
- The eight-dimensional dispersive constant.

Appendix A. The twenty-one layers

#	layer	checks	content
1	carrier	36	K5, D, spectrum, Lagrangian, exact motion, Petersen, Hodge block, edge tick, harmonic quartics
2	spectral	9	heat trace, ζ_D, spectral dimension, moments
3	spacetime	34	RP^2 pair, clock 125 → 120, E8 roots, walks, symbol, Green, clique
4	native	19	source, port and bath action; W_AS; 40/9; radiation channel; nonlinear flow
5	observer	7	readouts, dark sector, Witten index −5
6	gravity	8	Lin–Lu–Yau curvature; spectral action (CONDITIONAL)
7	cosmology	7	FLRW with the carrier field (CONDITIONAL)
8	frontier	23	band-edge orbit 135, the 224 saddle, saddle law, collar survey, coil
9	joins	24	the maps that make the layers one object
10	literal	20	the recovered operand file, every identity replayed
11	glue	11	E8 = D8 + s; A4 ⊕ A4′ with x ↦ ±2x
12	binding	12	DYN_BIND constraints
13	gates	12	TIME_BIND courts, register, W_L, T3081, AUTH package
14	recurrence	16	W_{90} exact, W_{600} mod p, exclusion, the order-16 operator

#	layer	checks	content
15	court	19	Court U19.81 audited
16	handover	30	the Locked Memory handover audited; critical census; T_6|V_c
17	waves	18	sound, light, frequency and phase on the K5 edges
18	snapshot	16	the 14:16 snapshot rebuilt; T^9 density; Hessians; asymptotics
19	spine	50	the circle lemma, the Coxeter clock, two carriers, the 7, what can act
20	canon	22	the canon read at 16:30 and 16:49; √30; the separate model replayed
21	ledger	19	the tick bit; the clock group; the five day clocks; golden; mirrors; carrier exchange; energy; spin; the clock file

Appendix B. Replay

```
pip install -r requirements.txt
python verify.py              # 21 layers, 412 checks, about 4 minutes
python -m pytest -q           # 31 tests
node --experimental-strip-types --no-warnings ports/ts/run.mjs   # the two TypeScript audits
```

The sibling's build was read from the Drive folder "One Machine 20261028": `one-machine-20261028-full.typ` / `.pdf`, `FULL_UPDATE_20261028.md` and forty join notes, 17:01–18:17 UTC. The canon was read from the Quick Access workbook, sheet 00_FULL_RECONCILIATION_20261028.